# *TECHNICAL SPECIFICATION AND DESIGN REPORT: OPTIMIZED STRUCTURAL MOUNTING BRACKET*

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## 1. INTRODUCTION & COMPONENT FUNCTION

This technical report details the geometric modeling and structural optimization of a multi-flanged mounting bracket designed for mechanical assembly and load distribution. The primary objective is to create a functional component capable of securing mechanical subsystems while minimizing structural stress concentration under load.

## 2. GEOMETRIC DESIGN SPECIFICATIONS

The component was modeled using Autodesk Tinkercad, incorporating specific spatial dimensions optimized for standard clearance fastening:

- Base Plate: Engineered for flat-surface structural anchoring with integrated precision clearance holes.
- Upright Wall: Vertical flange designed to manage multi-axis shear and bending forces.
- Fastener Holes: Standardized 6mm diameter circular clearance holes designed to accommodate mechanical bolts without localized material cracking.
- Reinforcement Gusset: Integrated a diagonal internal structural rib (gusset) to bridge the 90-degree corner joint, effectively shifting the moment of inertia and dispersing bending stress.

## 3. ADDITIVE MANUFACTURING & STRESS CONSIDERATIONS

When transitioning this digital CAD model to physical production via Fused Deposition Modeling (FDM) 3D printing, structural integrity depends heavily on internal slicing parameters:

- Infill Density & Pattern: Recommending a grid or groin infill structure at 40%+ density to balance material weight savings with high compressive load resistance.
- Print Orientation: Printing the component flat along the base plate to maximize layer adhesion shear strength along the primary load-bearing vector.

## 4. CONCLUSION

This model demonstrates the application of computer-aided design tools to solve mechanical joint challenges, proving that geometric reinforcement and thoughtful spatial alignment are critical to extending component fatigue life.